Internet of Things: Protocols and Networks

(CSC2106)

Bluetooth

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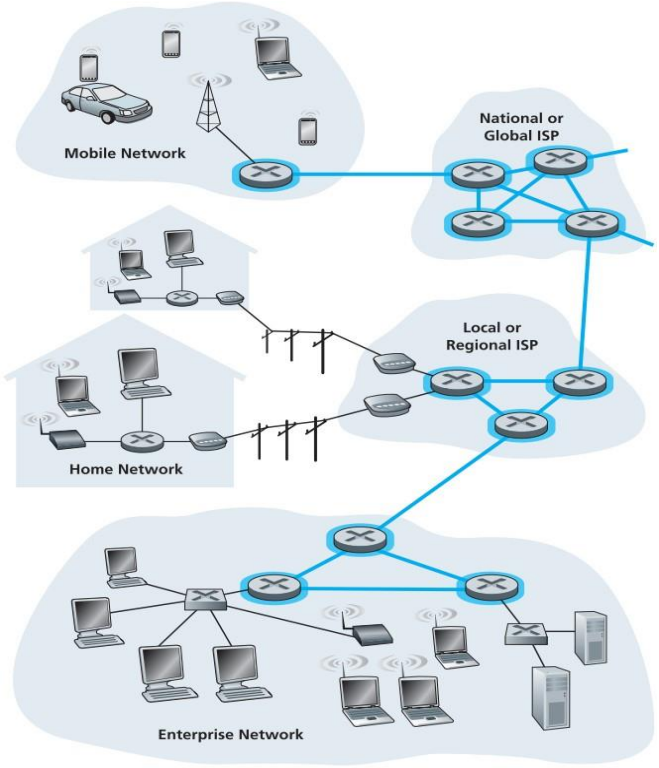
INF1006… Recap

- **What is Internet?**
  - Basically, an interconnection of computer networks, or packet-switching networks, or simply a network of networks.

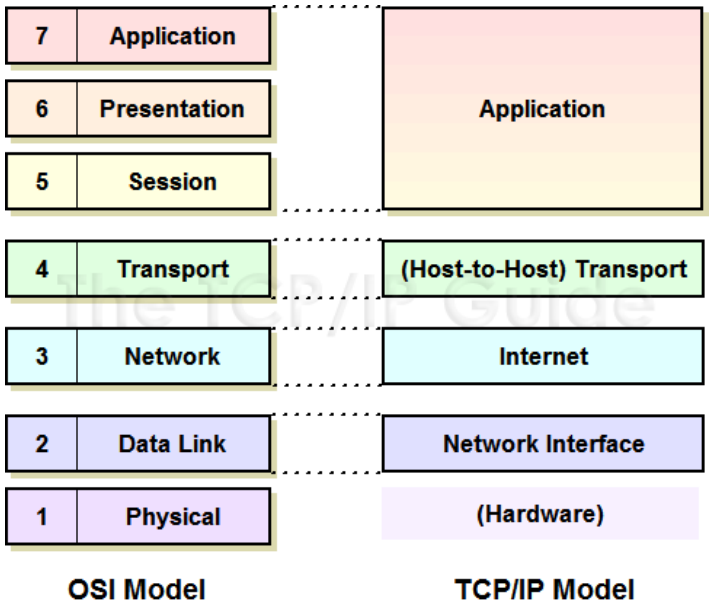


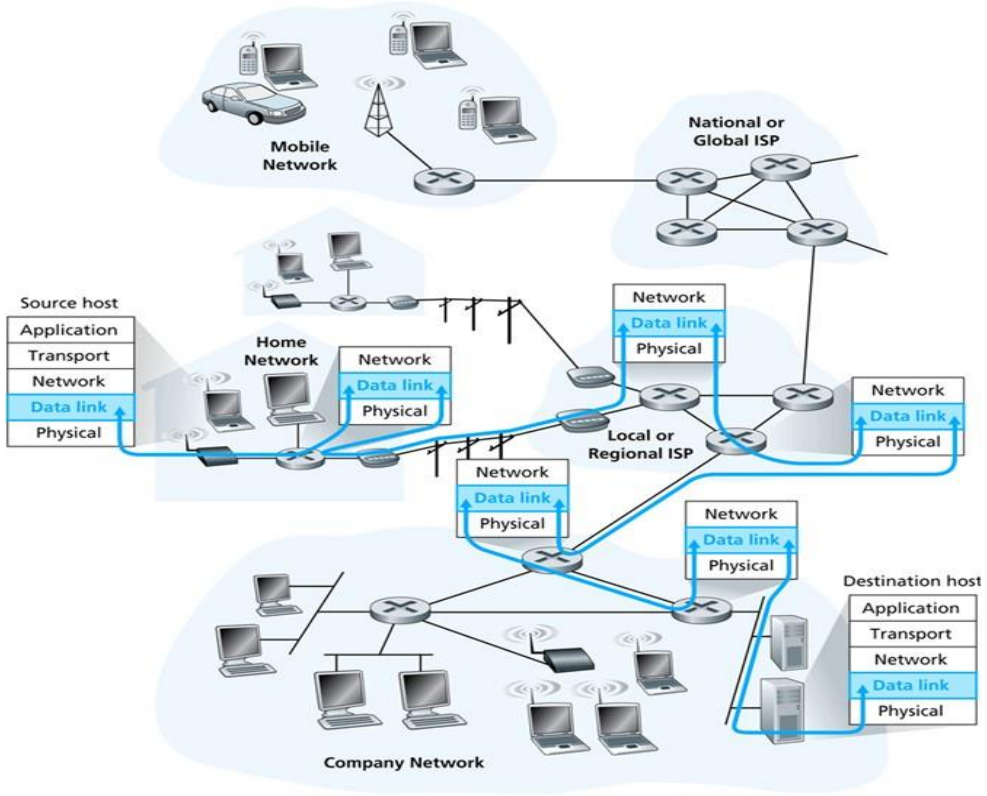
Network core: ISPs consisting of mesh of routers (packet-switches) and links.

Network edge: access networks, e.g. mobile, home and enterprise networks connected to the first router in the Internet.

INF1006… Recap

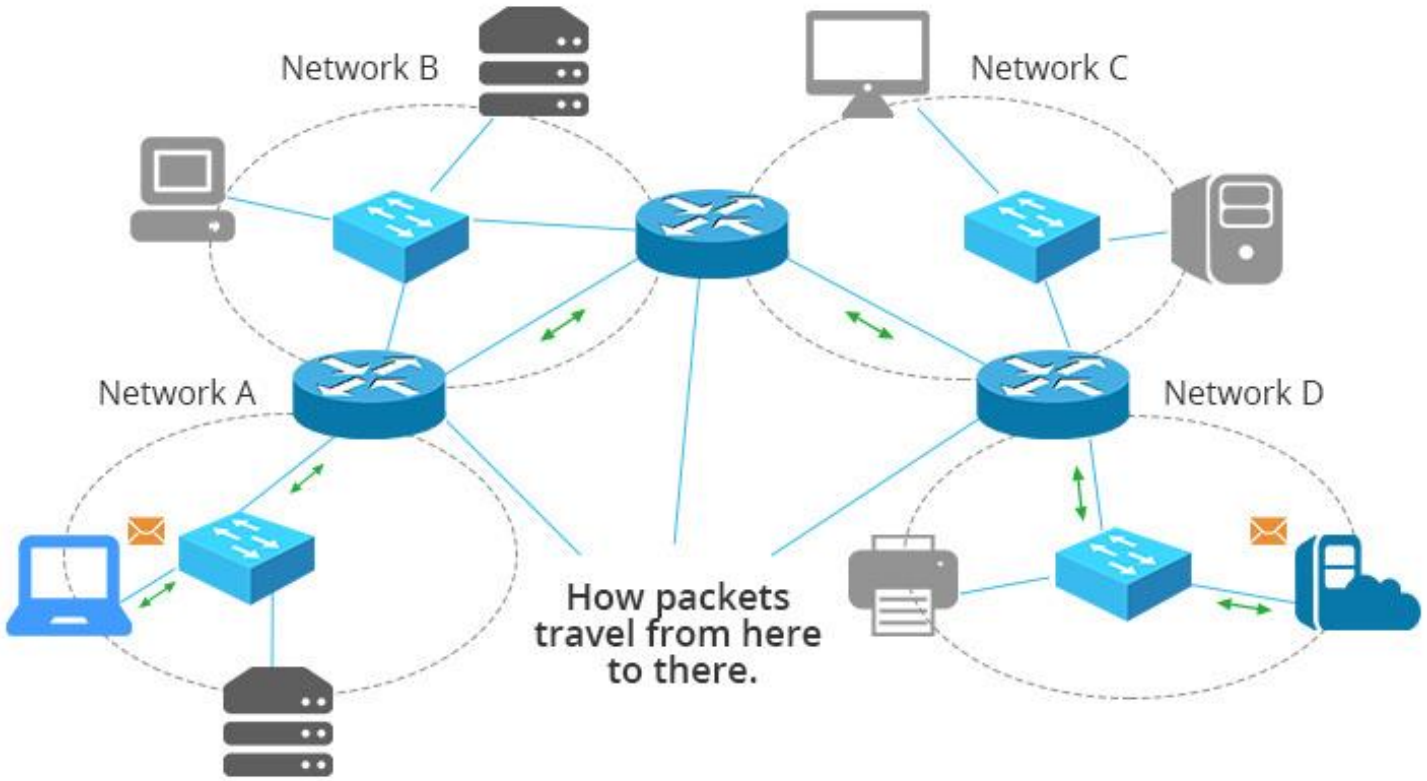
- **Layered Network Architecture Model**
  - The rationale for a layered network architecture model is to break down the big complex networking problem into small manageable parts, called layers in computer networks.

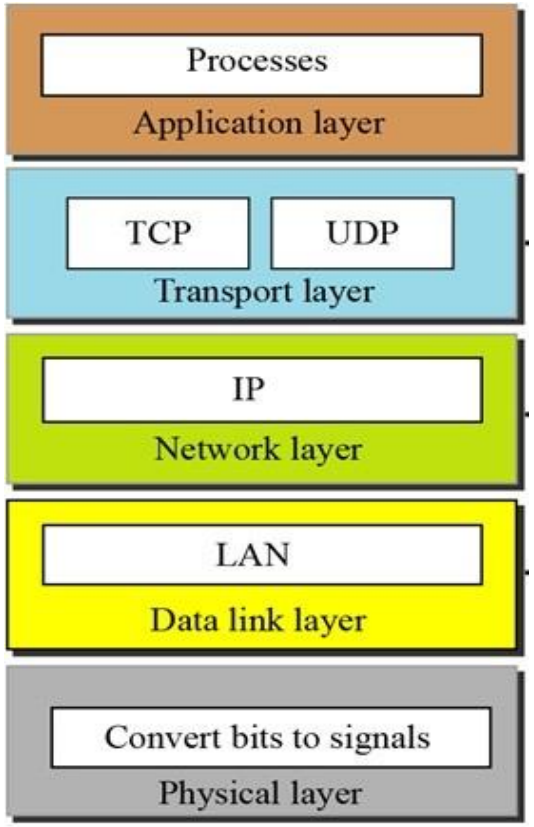




INF1006… Recap

Specifically for Internet to work:



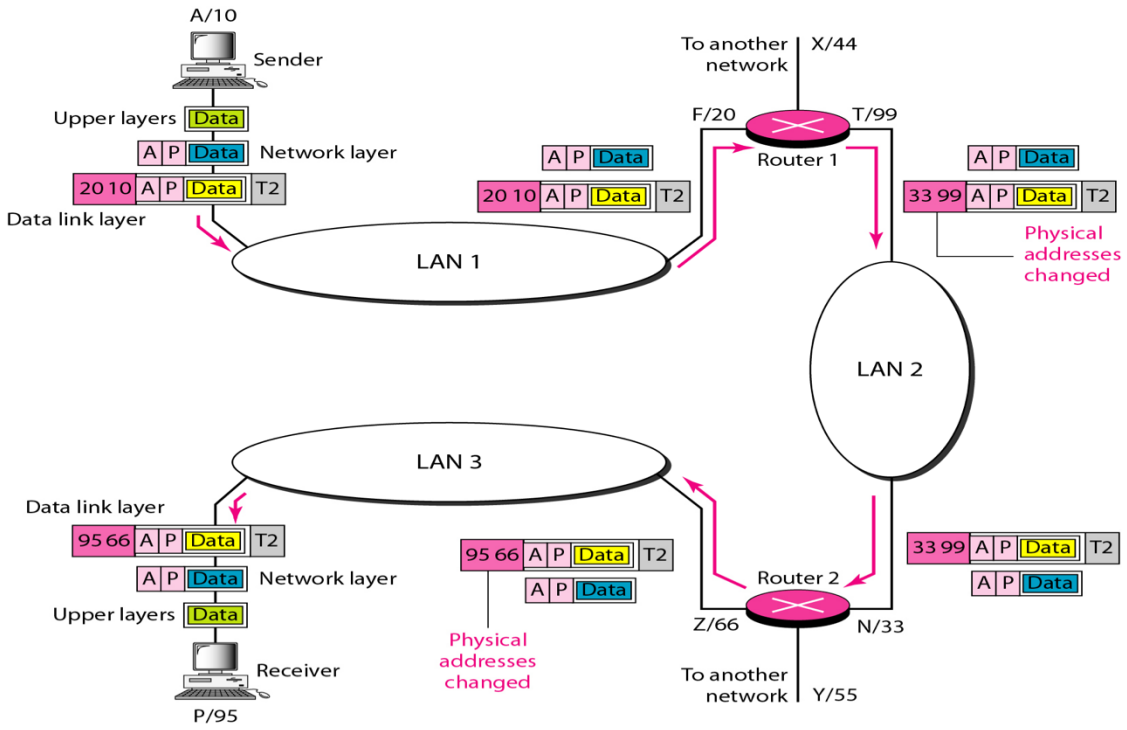


INF1006… Recap

- TCP/IP Communication Process
  - Working together, the task of each layer is in fact to perform **encapsulation** which is to add header (and trailer for data link layer) to higher layer data.
  - Finally, data will travel hop-by-hop from sender to intermediate nodes and eventually to the receiver which will perform **de-encapsulation**.

INF1006… Recap

- Re-encapsulation
  - Data will travel hop-by-hop from sender to intermediate nodes, which will perform de-encapsulation and **re-encapsulation** at each node, and eventually to the receiver.



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Such a nice protocol is available! Why do we need a different one for IoT? What is your thought?

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Coming back to the topic… IoT Ecosytem

Application	Smart Health, Smart Home, Smart Grid, Smart Transport, Smart Workspaces, …
Session	MQTT, CoRE, DDS, AMQP , …
Development Framework	Mbed, Homekit, AllSeen, IoTvity, ThingWorks, EVRYTHNG , …
Network	Encapsulation, 6LowPAN, 6TiSCH, Routing RPL, CORPL, CARP
Datalink	WiFi, Bluetooth Smart, ZigBee Smart, Z-Wave, DECT/ULE, 3G/LTE, NFC, Weightless, HomePlug GP, 802.11ah, 802.15.4, G.9959, WirelessHART, DASH7, ANT+ , LoRaWAN, …
Operating System	Linux, Android, Contiki-OS, TinyOS, …
Hardware	ARM, Arduino, Raspberry Pi, ARC-EM4, Mote, Smart Dust, Tmote Sky, …

Ref: Tara Salman, Raj Jain, "A Survey of Protocols and Standards for Internet of Things," Advanced Computing and Communications, Vol. 1, No. 1, March 2017, http://www.cse.wustl.edu/~jain/papers/iot_accs.htm

Recent Protocols for IoT

Session	MQTT, SMQTT, CoRE, DDS, AMQP , XMPP, CoAP, IEC,…	**Security**	**Management**
Network	Encapsulation 6LowPAN, 6TiSCH, 6Lo, Thread…	IEEE 1888.3, TCG, Oath 2.0, SMACK, SASL, EDSA, ace, DTLS, Dice, …	IEEE 1905, IEEE 1451, IEEE 1377, IEEE P1828, IEEE P1856
	Routing RPL, CORPL, CARP		
Datalink	Wi-Fi, 802.11ah, Bluetooth Low Energy, Z-Wave, ZigBee Smart, DECT/ULE, 3G/LTE, NFC, Weightless, HomePlug GP, 802.15.4e, G.9959, WirelessHART, DASH7, ANT+, LTE-A, LoRaWAN, ISA100.11a, DigiMesh, WiMAX, **NB-IoT**, SigFox …		

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So many protocols! Do you think we need so many? Why can't we have only one like TCP/IP?

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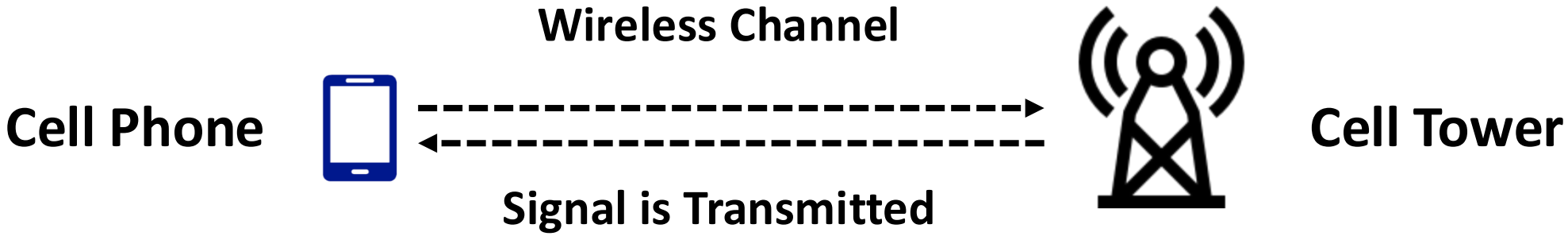
IoT protocols

Table 1. Comparison of protocols

Protocol	Frequency	Distance	Data Transfer Speed
Bluetooth	2.4 GHz	50-150 m	1 Mbps
ZigBee	2.4 GHz	10-100 m	250 kbps
Z-Wave	900 MHz	30 m	9.6 / 40/100 kbps
6LoWPAN	2.4 GHz	10-100 m	250 kbps
NFC	13.56 MHz	10 cm	100-420 kbps
Sigfox	900 MHz	30-50 km in the countryside, 3-10 km city center	10-1000 bps
LoRaWAN	Unstable	2-5 km, 15 km	0,3-50 kbps

Ref: Aslan, Füsun Yavuzer, and Bora Aslan. "Comparison of IoT Protocols with OSI and TCP/IP Architecture." *International Journal of Engineering Research and Development* 15, no. 1: 333-343.

Preliminaries



Messages are represented using bits ‘1’ and ‘0’.
For example, APPLE in binary: **0100000101010000010100000100110001000101**.

Note that the messages are not transmitted as it is.

Preliminaries

- A group of bits are **encoded** as a symbol before transmitting over a wireless medium
  - Example 1: **Binary Phase Shift Keying (BPSK)**
    - **BPSK**: the symbol for bit ‘1’ is ‘+1’ and for bit ‘0’ is ‘-1’

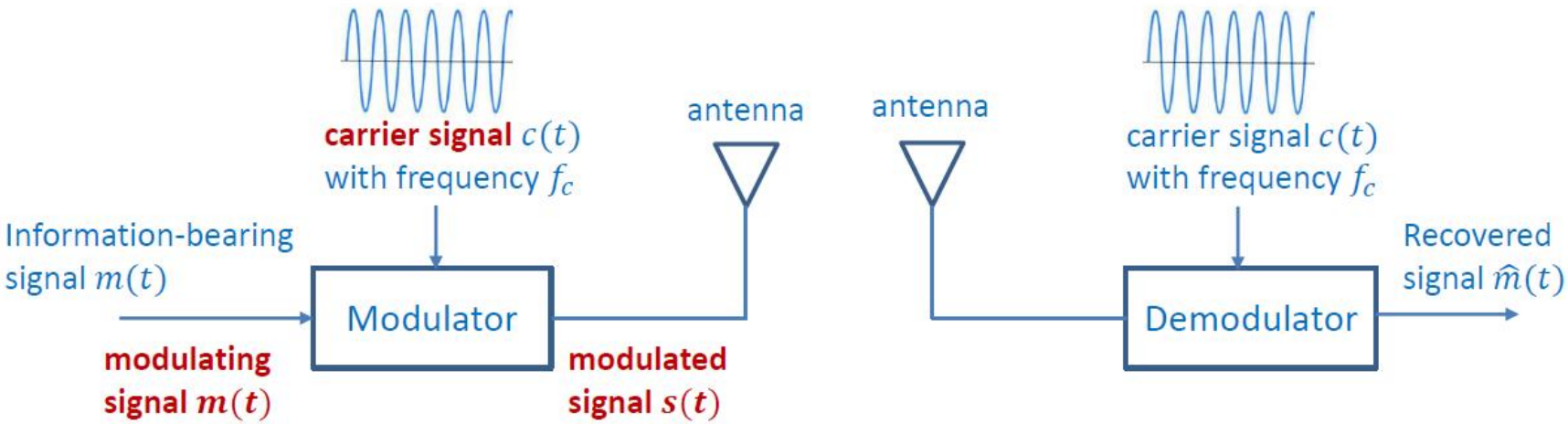
0	1	0	0	1	0	0	1	→	-1	+1	-1	-1	+1	-1	-1	+1

  - Example 2: **Quadrature Phase Shift Keying (QPSK)**
    - **QPSK:** for every two bits one symbol is assigned

- Further, to generate the signal required to be transmitted, the symbols are multiplied by a carrier, a cosine function generally represented as $\cos(2*pi*f_c t)$ where f_c is the carrier frequency.

Preliminaries

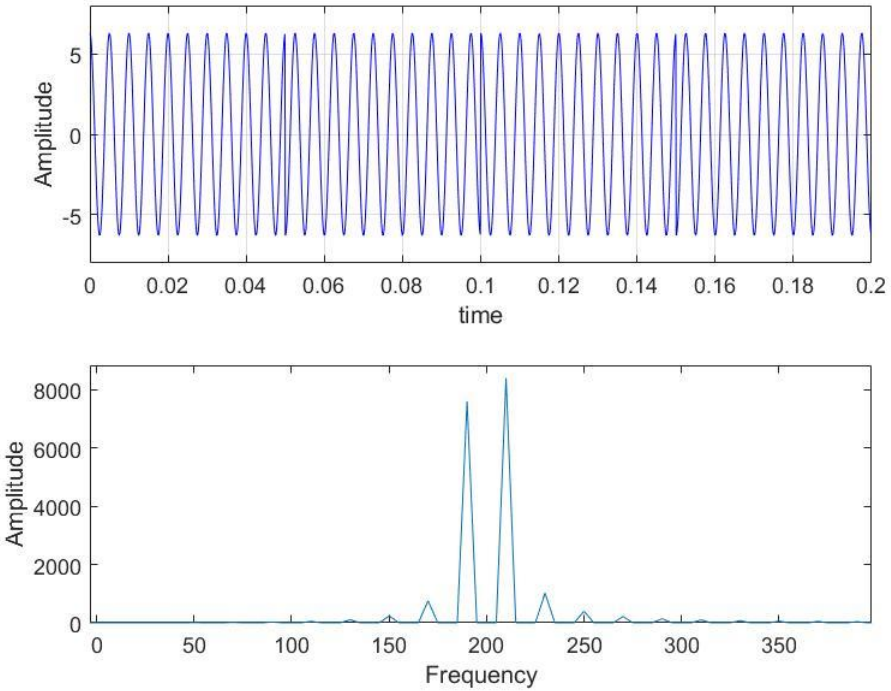
- How single users send data?



Preliminaries

- **Wireless Signal:**
  - Can be represented in time domain and frequency domain

- Need for Multiple Access?
  - Yes, for multiple users to transmit
  - Will be used in subsequent lectures
  - Like CSMA/CD discussed in INF1006
  - Different Multiple Schemes:
    - Time Division Multiple Access (TDMA)
    - Frequency Division Multiple Access (FDMA)
    - Code Division Multiple Access (CDMA)
    - Orthogonal Frequency Division Multiple Access (OFDMA)



Bluetooth

- Started with Ericsson's Bluetooth Project in 1994 for radio-communication between cell phones over short distances
- Named after Danish king Herald Blatand (AD 940-981)[1]
- Intel, IBM, Nokia, Toshiba, and Ericsson formed Bluetooth SIG in May 1998
- Version 1.0A of the specification came out in late 1999
- IEEE 802.15.1 approved in early 2002 is based on Bluetooth. Later versions handled by Bluetooth SIG directly
- Key Features at **time of launch** in 1999-2000:
  - Lower Power: 10 mA in standby, 50 mA while transmitting
  - Cheap: $5 per device
  - Small: 9 mm2 single chips

1. https://www.bluetooth.com/about-us/bluetooth-origin/#:~:text=Surprisingly%2C%20the%20name%20dates%20back,earned%20him%20the%20nickname%20Bluetooth.

Bluetooth Versions

Bluetooth®

1999
Bluetooth 1.0

- First formal Version
- 1.1 point-to-multipoint
- 1.2 adaptive frequency hopping

2004
Bluetooth 2.0

- Enhanced data rate design
- 2.1 Secure paring/connection

2009
Bluetooth 3.0

- Speed up to 24Mbps
- Transfer through WiFi channel

2010
Bluetooth 4.0

- Low Energy design
- 4.1 support for LTE
- 4.2 support IPv6 and 6LoWPAN

2016
Bluetooth 5.0

- Long range PHY
- 5.1 location services (AoA/AoD)
- 5.2 LE audio, EATT
- 5.3 advertising enhancements
- 5.4 periodical advertising w/response

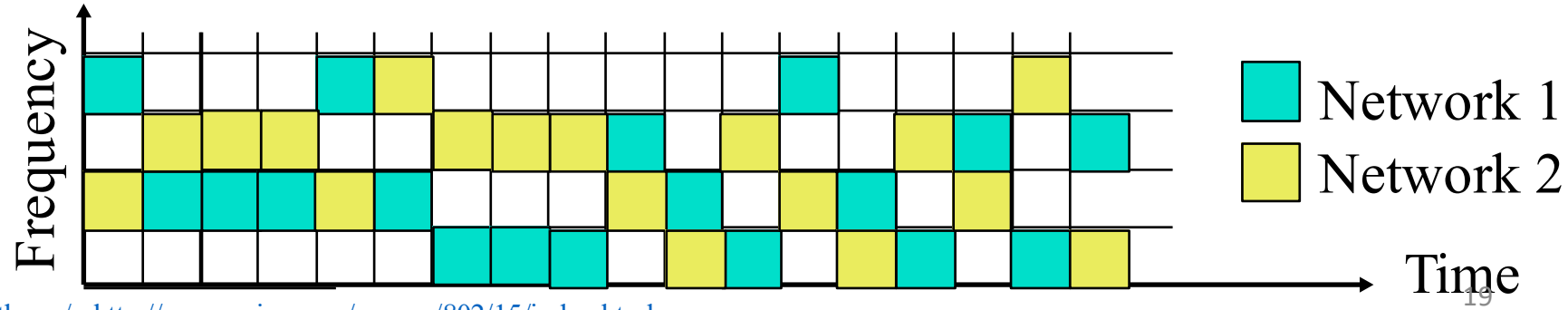
2024
Bluetooth 6.0

- Bluetooth Channel Sounding
- Decision-Based Advertising Filtering and Monitoring Advertisers
- ISOAL and timing improvements

Version group	Main focus of improvement	Typical effect
1.0–2.x	Basic wireless cable, then higher data rate and better pairing	Short-range audio and peripherals
3.0	High speed via Wi-Fi	Faster file transfers
4.x	Low Energy (BLE)	Years of battery life for small sensors
5.0–5.3	Range, speed, direction finding, LE Audio	Industrial IoT, RTLS, richer audio
5.4	Massive low-power broadcast fleets	ESL, large sensor networks
6.0	Precise distance, smarter scanning, lower latency	Digital keys, high accuracy tracking, denser deployments

Bluetooth: Details

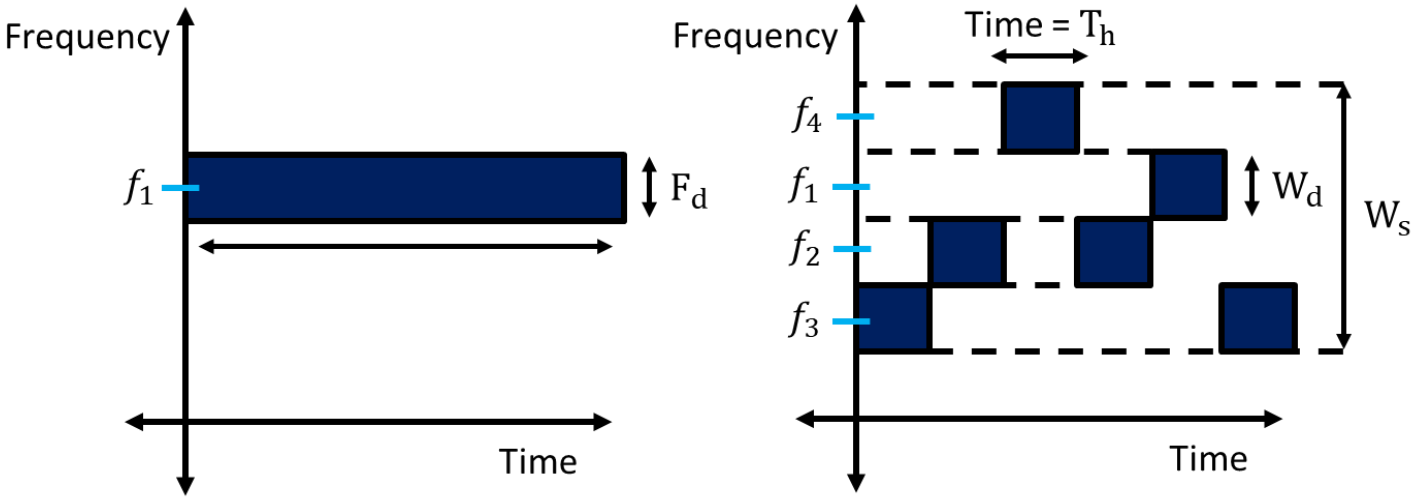
- **Frequency Range:** 2402 - 2480 MHz
  - Total 79 MHz band
  - 23 MHz in some countries, e.g., Spain, France and Japan
- **Data Rate:** 1 Mbps using 1 MHz (Nominal) 720 kbps (User)
- **Enhanced Data Rate:** Possible to achieve 3Mbps with better encoding
- **Frequency Hopping Spread Spectrum:** 1600 times/s → 625 us/hop (microseconds/hop)
  - Hopping Sequence decided using PRN generator agreed upon between master and slave
- **Security:** Challenge/Response Authentication. 128b Encryption
- **TX Output Power:**
  - Class 1: 20 dBm Max. (0.1W) – 100m
  - Class 2: 4 dBm (2.5 mW)
  - Class 3: 0 dBm (1mW) – 10m



Ref: http://www.bluetooth.com/, http://www.bluetooth.org/, http://grouper.ieee.org/groups/802/15/index.html

Frequency Hoping Spread Spectrum (FHSS)

- FHSS: The type of spread spectrum in which carrier hops randomly from one frequency to another.



- Symbol definitions:
  - T_h → Hopping Time, W_d → Signal Bandwidth, W_s → Spread Spectrum bandwidth, C → number of carrier frequencies (or channels allocated for the FH signal)
  - Note that $W_s = CW_d$

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To test if you understood it right: If the signal bandwidth is 10MHz and the number of carrier frequencies is equal to 20, what is the spread spectrum bandwidth?

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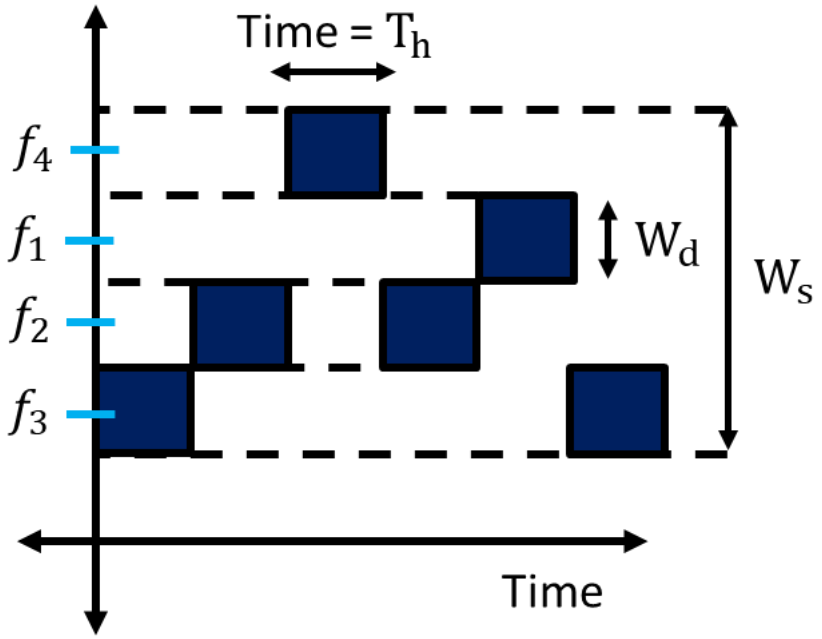
Frequency Hoping Spread Spectrum (FHSS)

- Channel sequence, $\{f_3,\ f_2,\ f_4,\ f_2,\ f_1,\ f_3,\ \cdots\}$, is a random series of radio frequencies.
- The sequence of channels used is dictated by a predefined random number (seed).
- In the example shown, the number of carrier frequencies i.e., C is equal to 4.
- Note that both transmitter and receiver use the same seed to tune into a sequence of channels in synchronization.
- The transmitter operates in one channel at a time for a fixed interval
- At each successive interval, a new carrier frequency is selected



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Device 1 is sending a bluetooth signal with the Channel sequence {f1, f2, f4, f7, f3}. The channel sequences at other devices are as follows:

Device 2: {f1, f5, f7, f4, f3}
Device 3: {f1, f2, f4, f7, f9}
Device 4: {f1, f2, f4, f7, f3}
Device 5: {f1, f7, f3, f4, f2}

Which device will be able to successfully decode Device 1's data?

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Generating the Frequency Sequence

Produces an output Pn_i at i^{th} time instant

Produces an output f_i at i^{th} time instant

The output is equal to $\cos(2\pi f_i t)$

Pseudo Random Number (PRN) → Channel Table (Centre Frequency) → Frequency synthesizer → Output

At a given time instant, this generates k-bit number i.e., the output is k bits

PRN	Frequency
$00 \cdots 00$	f_1
$00 \cdots 01$	f_2
$\vdots$	$\vdots$
$11 \cdots 11$	f_C

This produces the corresponding cosine output

- Point to be Noted:
  - Using a k-bit PRN generator, we can have a maximum of 2^k carrier frequencies or channels.

Bluetooth – Channel Acess

- Frequency Hopping Spread Spectrum
  - Total bandwidth divided into 1MHz physical channels
  - Hopping sequence shared with all devices on piconet
  - Maximum Radio Frequency Hopping:
    - 1600 times per second
    - Minimum Hop Time is equal to 625 microseconds
    - Slot Time is equal to 625 microseconds
  - Packets = 1 slot, 3 slot, or 5 slots long
  - The frequency hop is skipped during a packet.

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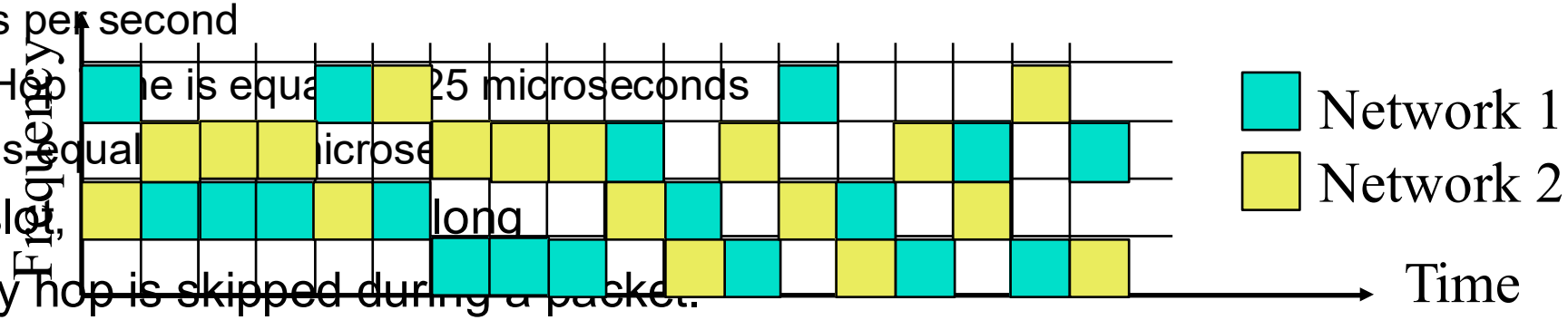
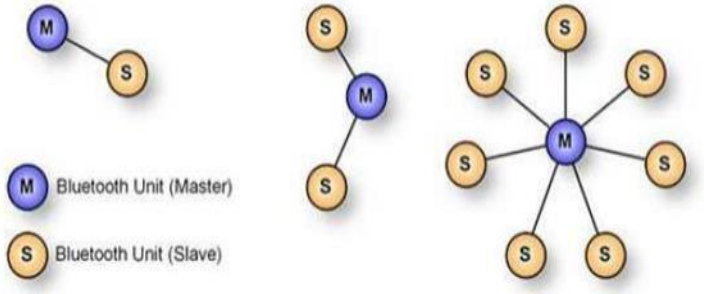


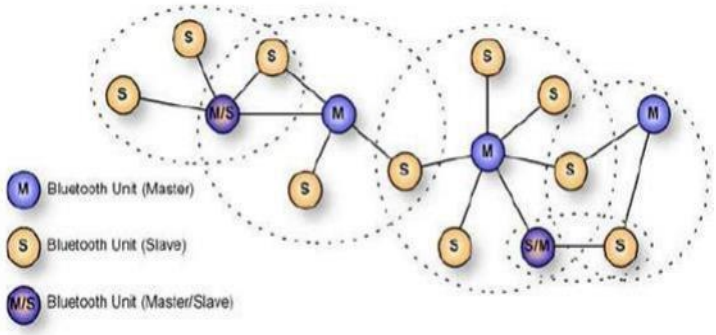
- TDD:
  - Master starts in even numbered slots only.
  - Slaves start in odd numbered slots only.

Topology



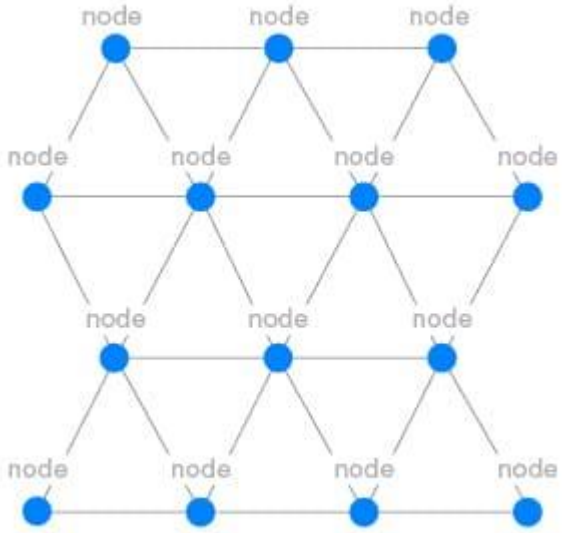
Piconet v4.0

Up to 7 active slave per master..
Up to 255 'parked' slaves per master



Scatter net v4.1

Multiple piconets joined to form Scatternet

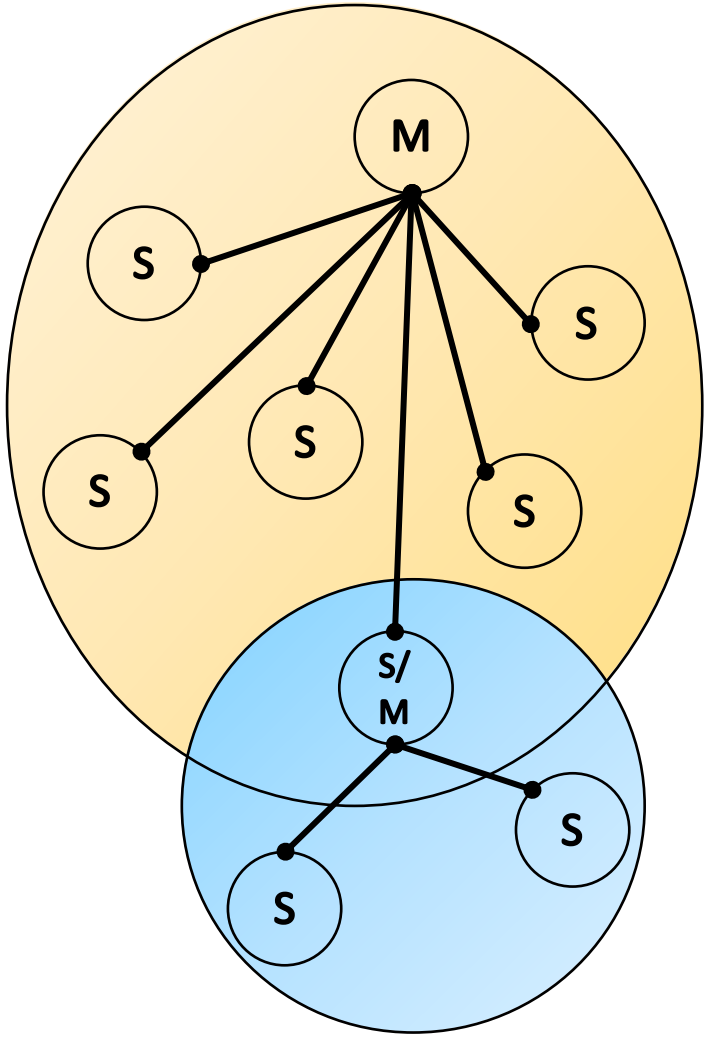


Bluetooth Mesh v5.0

Bluetooth Mesh: https://www.bluetooth.com/blog/the-fundamental-concepts-of-bluetooth-mesh-networking-part-1/

Topology contd …

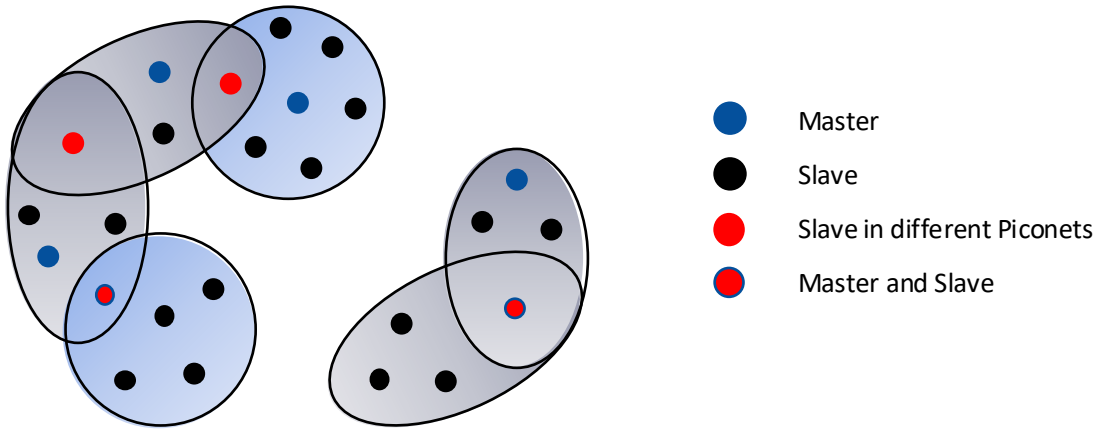
- Piconet is formed by a master and many slaves
  - Basic unit of Bluetooth networking
  - Up to 7 active slaves and up to 255 Parked slaves.
  - Slaves can only transmit when requested by master
- Active slaves are polled by master for transmission
- Each station gets an 8-bit parked address
  - 255 parked slaves/piconet
- The parked station can join in 2us. Other stations can join in more time.
- A Bluetooth node can be both a master in one piconet and slave in another.



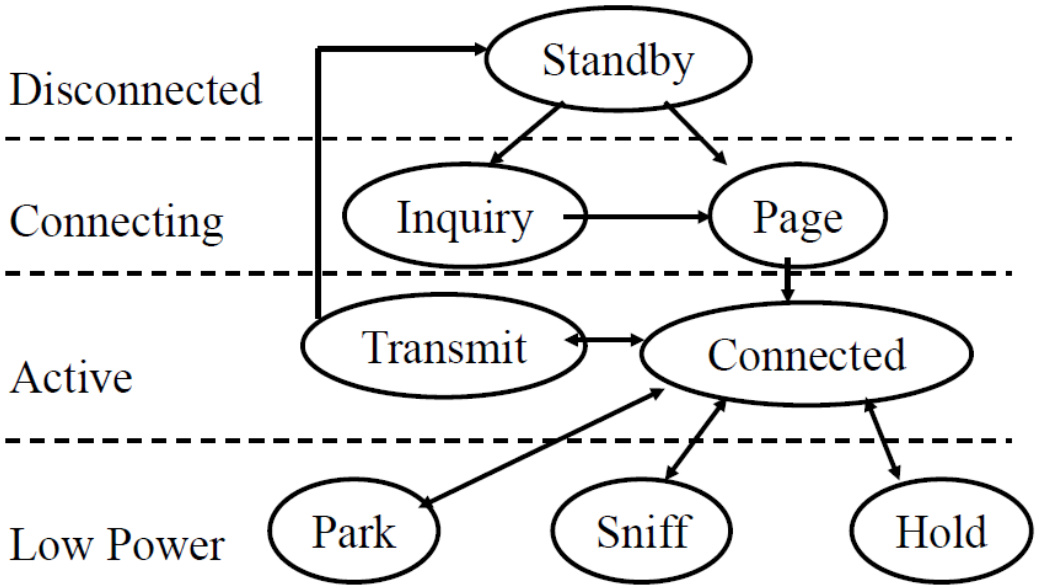
Ref: P. Bhagwat, "Bluetooth Technology for short range wireless Apps," IEEE Internet Computing, May-June 2001, pp. 96-103

Topology contd …

- Scatternet:
  - Device in one piconet can exist as master or slave in another piconet
  - Allows many devices to share same area
  - Makes efficient use of bandwidth
  - Main Disadvantage:
    - Collisions can occur when devices in different piconets use the same Frequency Hopping Sequence



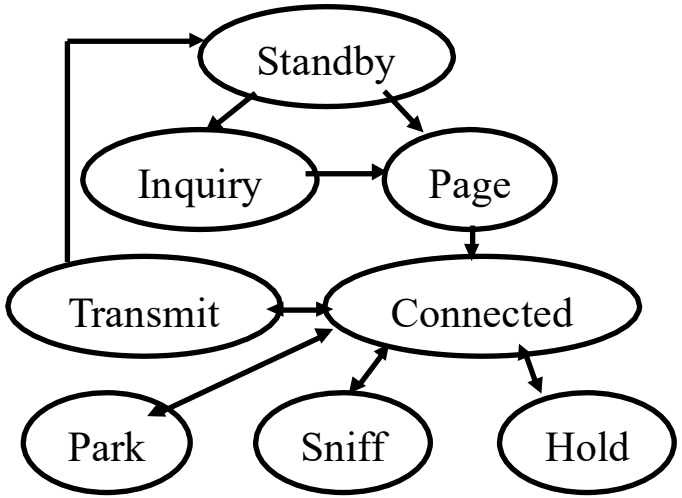
Bluetooth Operational States



- **Standby**: Initial state
- **Inquiry**: Master sends an inquiry packet. Slaves scan for inquiries and respond with their address and clock after a random delay (CSMA/CA – Carrier Sense Multiple Access/Collision Avoidance)

Bluetooth Operational States contd …

- **Page**: Master in page state invites devices to join the piconet. Page message is sent in 3 consecutive slots (3 frequencies). Slave enters page response state and sends page response including its device access code.
- Master informs slave about its clock and address so that slave can participate in piconet. Slave computes the clock offset.
- **Connected**: A short 3-bit logical address is assigned
- **Transmit**: Data is Transmitted

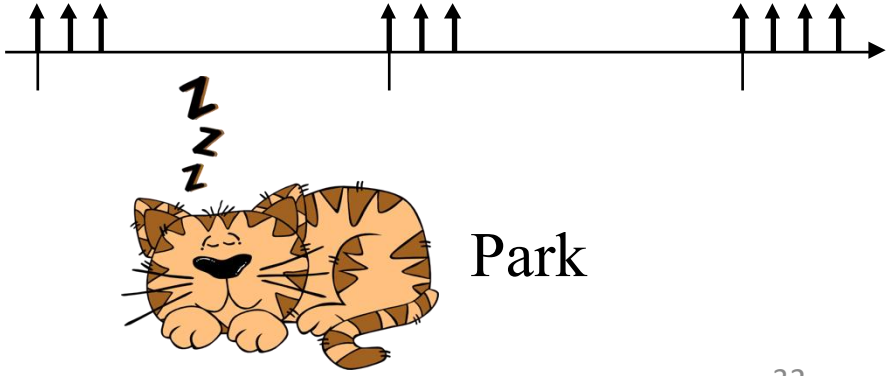


Energy Management in Bluetooth

Three inactive states:

1. **Hold**: No Asynchronous Connection List (ACL). Synchronous Connection Oriented (SCO) continues. Node can do something else: scan, page, inquire
2. **Sniff**: Low-power mode. Slave listens after fixed sniff intervals.
3. **Park**: Very Low-power mode. Gives up its 3-bit active member address and gets an 8-bit parked member address. Wake up periodically and listen to beacons. Master broadcasts a train of beacons periodically (has been removed in BLE 6 specification)

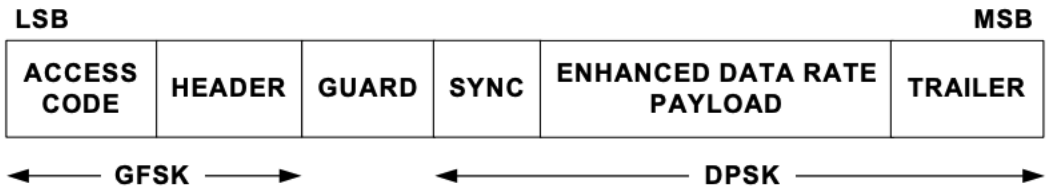
Sniff



Park

Bluetooth Frame Format (MAC)

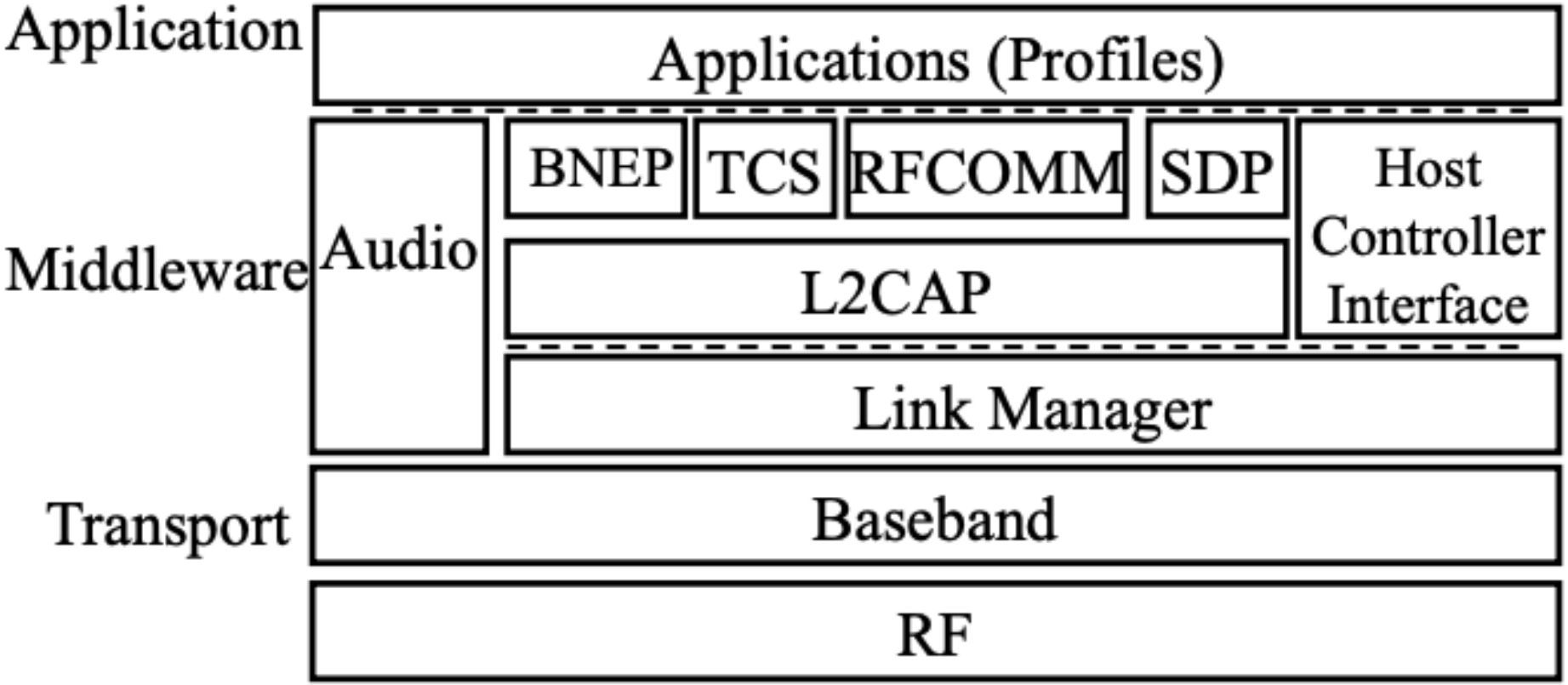
Access Code	Baseband/Link Control Header	Data Payload
72b	54b	0-2744b



- Frames can be up to five slots long. Five slots =3125 us.
- Access codes: (also indicates to the receiver the arrival of a packet. It is used for timing synchronization and offset compensation.)
  - Channel access code identifies the piconet (used in connection state)
  - Device access code for paging requests and response (used during paging)
  - Inquiry access code to discover units (used during inquiry)
- Header: member address, type code, flow control, ack/nack, sequence number, and header error check.
- Each device has a 48-bit IEEE MAC address: Lower address part (LAP) - 24 bits, Upper address part (UAP) - 8 bits and Non-significant address part (NAP) - 16 bits
  - UAP+NAP = Organizationally Unique Identifier (OUI) from IEEE

Ref: https://www.bluetooth.com/specifications/specs/core-specification-3-0-hs/
https://www.mathworks.com/help/bluetooth/ug/bluetooth-packet-structure.html#mw_6f94d3a3-12d3-450b-8c1d-55696a4e6ad6

Bluetooth Protocol Stack



Ref: https://www.cse.wustl.edu/~jain/cse574-24/ftp/j_11blez.pdf

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Quiz Time!

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What changes to be done to make the Bluetooth protocol useful for IoT?

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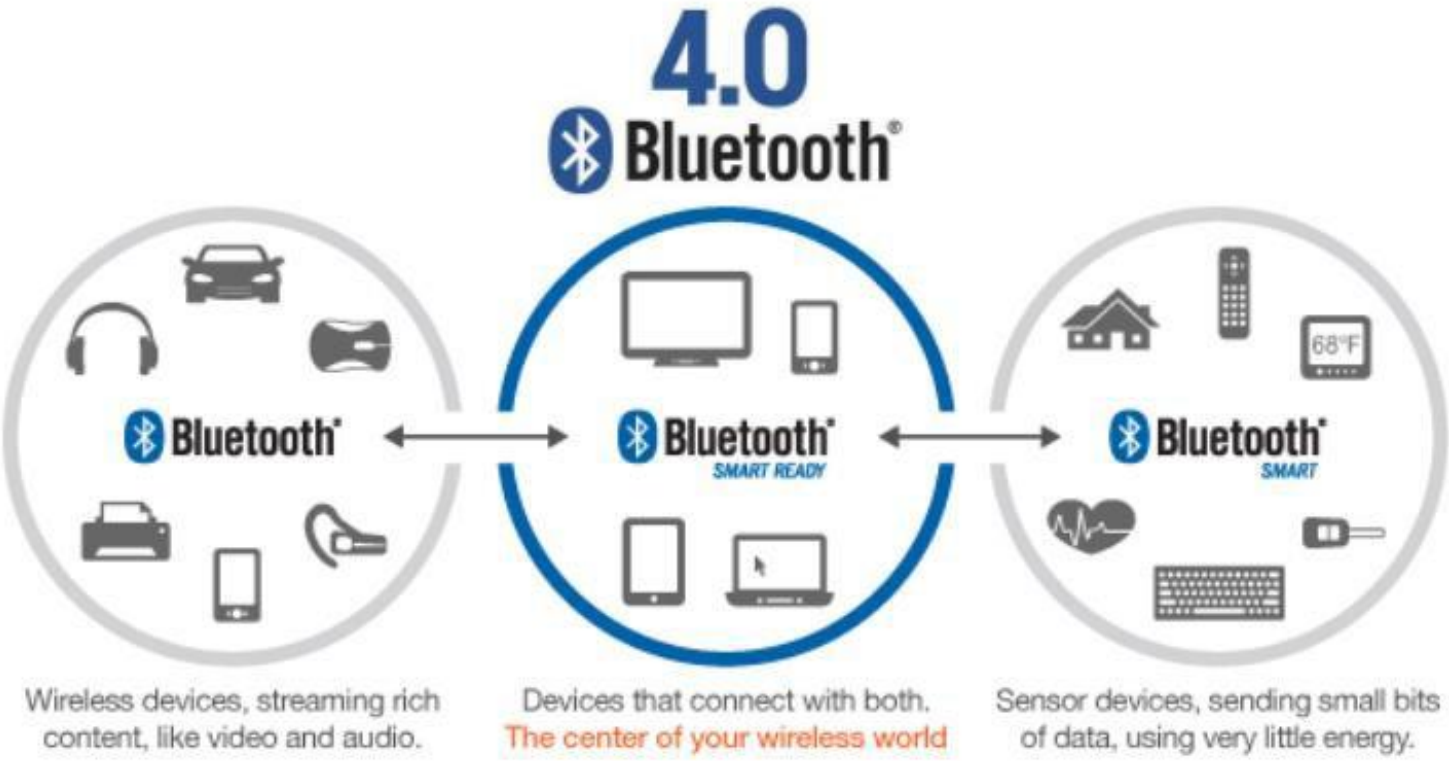
Bluetooth 4.x and beyond

- Bluetooth 4.0
- Bluetooth Low Energy
  - BLE, BTLE, LE
- SIG Preferred
  - Bluetooth Smart
  - Bluetooth Smart Ready



Bluetooth Smart or BLE

- **Low Energy**: 1% to 50% of Bluetooth classic
- **For short broadcast**: Your body temperature, Heart rate, Wearables, **sensors**, automotive, industrial

  Not for voice/video, file transfers, …
- **Small messages**: 1Mbps data rate but throughput not critical

Do not edit
How to change the design

Why is it that throughput is not critical? What are your thoughts?

The Slido app must be installed on every computer you're presenting from

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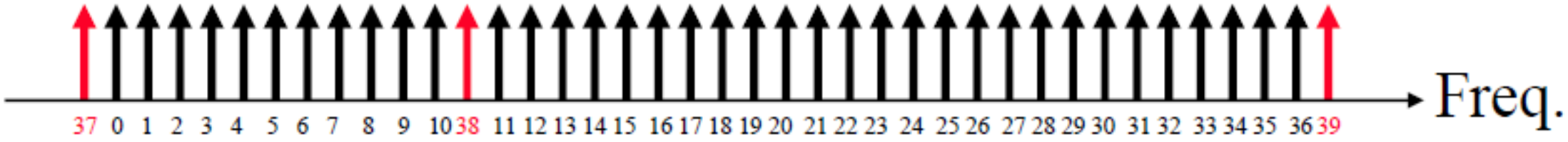
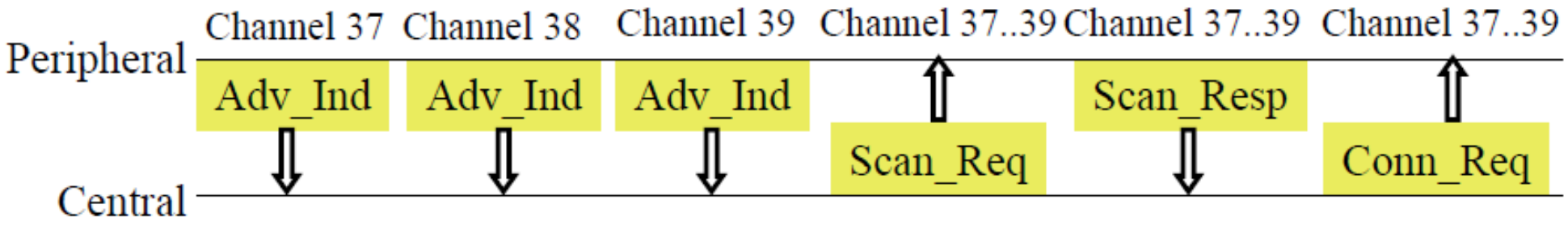
- **Small messages**: 1Mbps data rate but throughput not critical
- **Battery life**: In years from coin cells
- **Simple**: Star topology, Mesh (From BLE 5.1)
- **Lower cost** than Bluetooth classic
- New protocol design based on Nokia's **WiBree** technology. It shares the same 2.4GHz radio as Bluetooth
  - Dual mode chips (operates in old mode or the new mode)
  - All new smart phones (iPhone, Android, …) have dual-mode chips

BLE PHY

- Two Device Types: “**Peripherals**” simpler than “**central**”
- Transmits at lower power than Bluetooth Classic and has intelligent power management
- Lower Connection Time
- 1 Mbps data rate for BLE 4.2 and enhanced to 2 Mbps for BLE 5 and 6
- Adaptive Frequency hopping. **40 Channels with 2 MHz** spacing
  - 3 channels reserved for advertising and 37 channels for data
  - Advertising channels specially selected to avoid interference with Wi-Fi channels





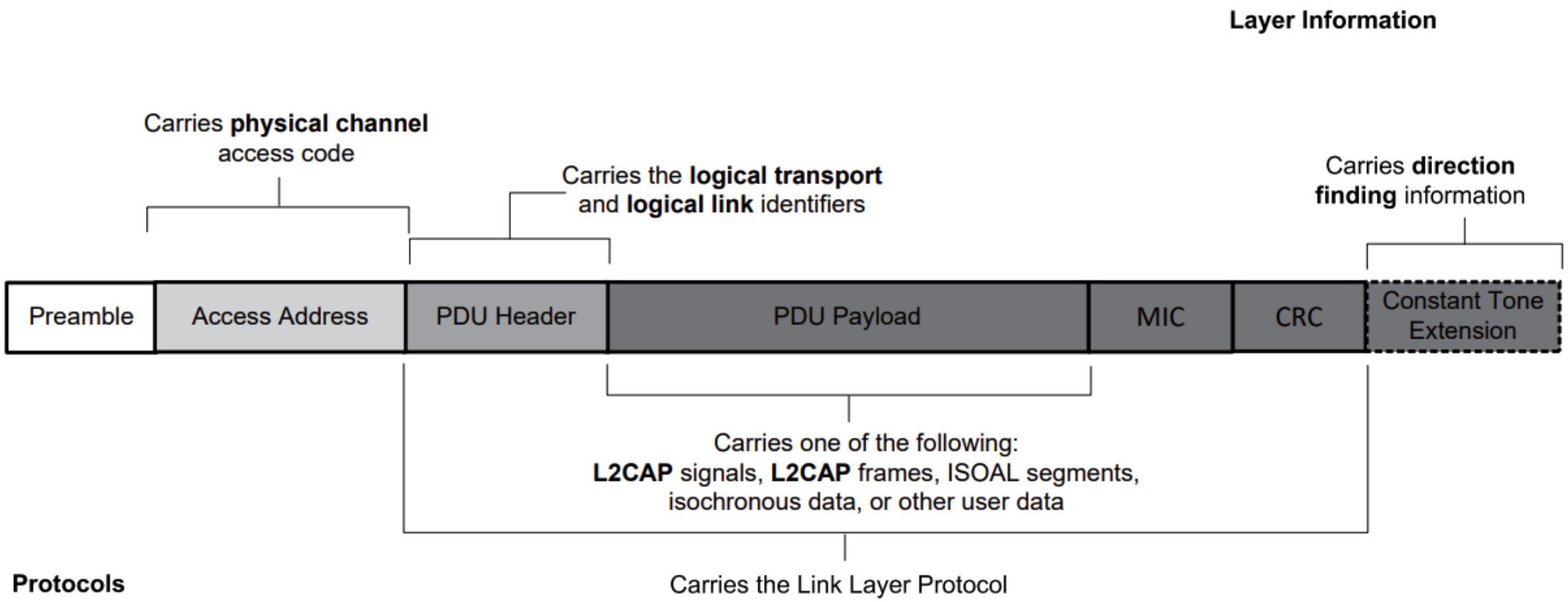
BLE Operational Procedures

- Device Filtering Procedure
- Advertising Procedure
- Scanning Procedure
- Discovering Procedure
- Connecting Procedure
- Connected Mode
- Periodic advertising Procedure
- Periodic advertising synchronization procedure
- Periodic advertising synchronized mode
- Decision-based scanning

The flexible operational procedures in BLE such as adjustable advertising behaviour, connection intervals, and sleep scheduling are a key reason for its low energy consumption.

Ref: https://www.bluetooth.com/specifications/specs/core-specification-6-0/ (Pages 303 to 308)

BLE MAC Packet Structure



Ref: https://www.bluetooth.com/specifications/specs/core-specification-6-0/

BLE Protocol Stack

ATT: It defines how data is represented in a BLE server database and the methods by which that data can be read or written.

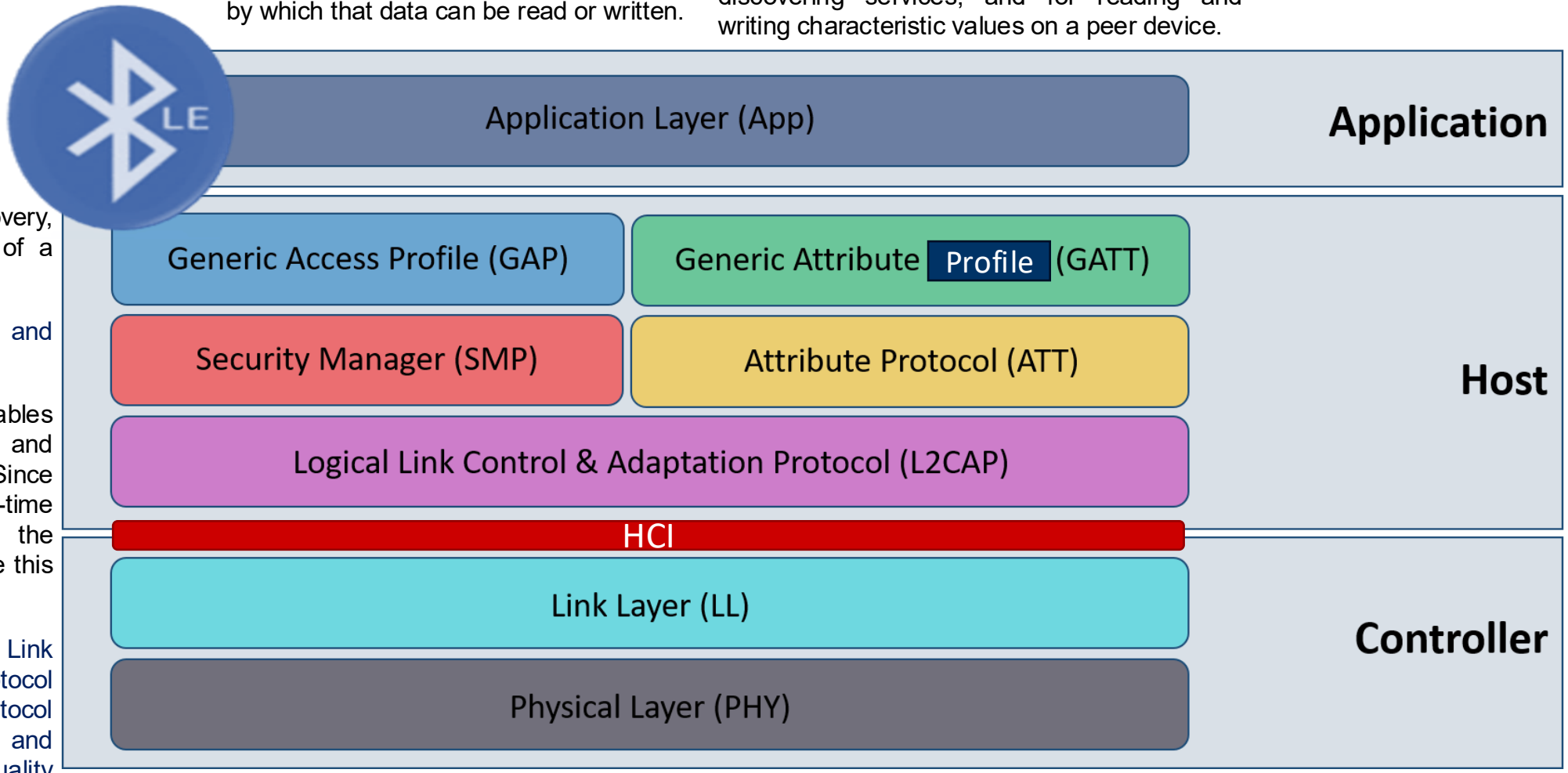
Generic Attribute Profile describes a service framework using the Attribute Protocol for discovering services, and for reading and writing characteristic values on a peer device.

GAP: responsible for device discovery, determines the network topology of a BLE system, security procedures

SMP: Pairing, Authentication and Encryption

Host Controller Interface (HCI) enables communication between the host and the controller via a serial interface. Since the controller deals with hard real-time requirements and contact with the physical layer, it is good to separate this part from the host.

L2CAP: The Bluetooth Logical Link Control and Adaptation Protocol (L2CAP) supports higher level protocol multiplexing, packet segmentation and reassembly, and the conveying of quality of service information.



PHY Layer takes care of transmission and receiving, modulation and demodulation, etc. 40 channels of 2MHz

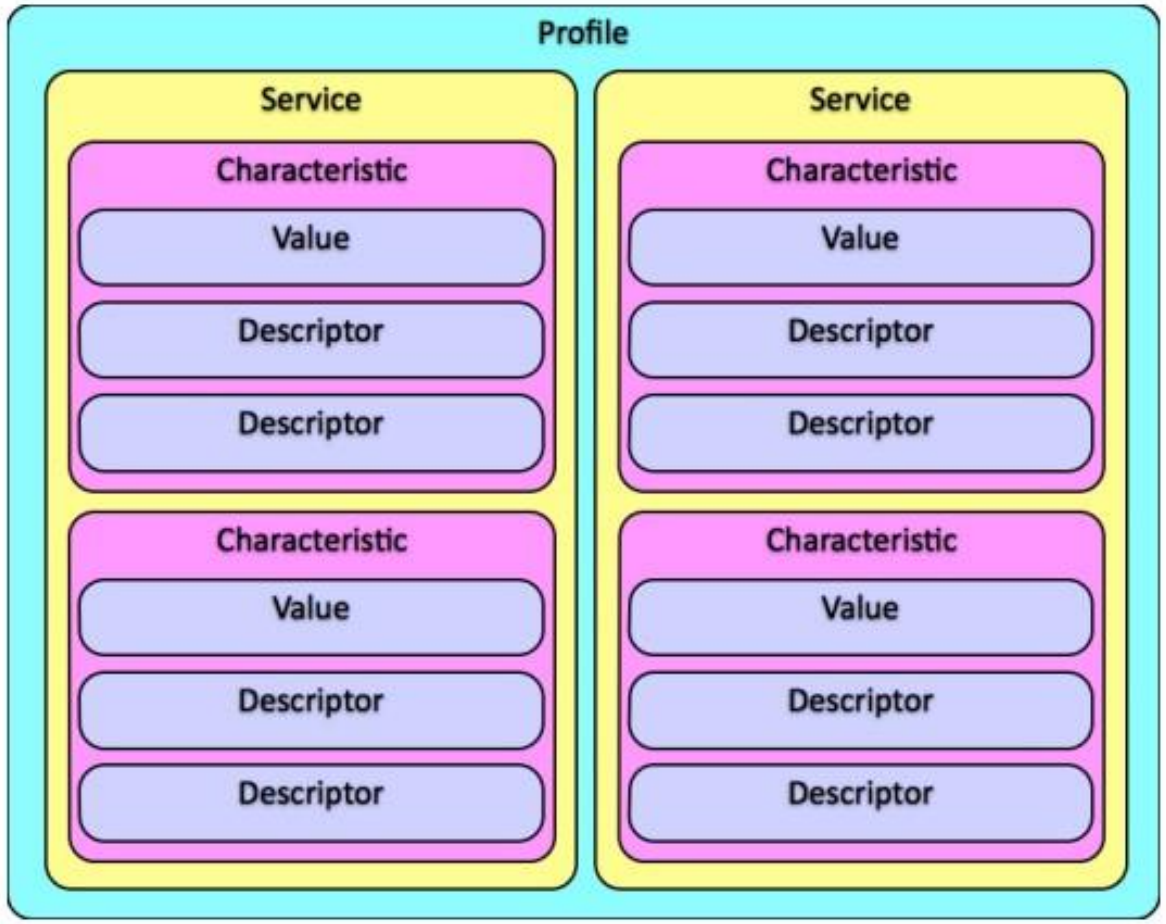
Link Layer defines packet structure and control

Attribute Protocol (ATT)

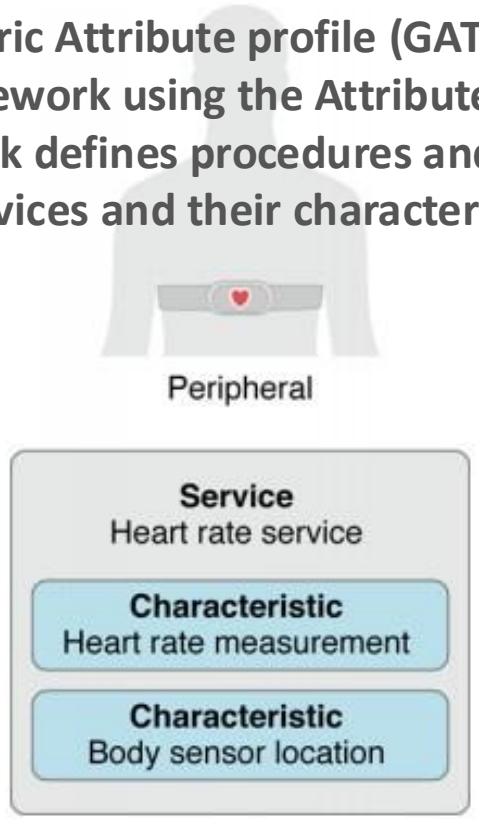
- The Attribute Protocol defines two roles; a server role and a client role. It allows a server to expose a set of attributes to a client that are accessible using the Attribute Protocol.
- An attribute is a discrete value that has the following three properties associated with it:
  - attribute type, defined by a UUID
  - attribute handle
  - a set of permissions that are defined by each higher layer specification that utilizes the attribute; these permissions cannot be accessed using the Attribute Protocol.

Type	Purpose
Commands	PDUs sent to a server by a client that do not invoke a response.
Requests	PDUs sent to a server by a client that invoke a response.
Responses	PDUs sent to a client by a server in response to a request.
Notifications	Unsolicited PDUs sent to a client by a server that do not invoke a confirmation.
Indications	Unsolicited PDUs sent to a client by a server that invoke a confirmation.
Confirmations	PDUs sent to a server by a client to confirm receipt of an indication.

Generic Attribute Profile - GATT



The Generic Attribute profile (GATT) defines a service framework using the Attribute Protocol. This framework defines procedures and formats of services and their characteristics.



Services, characteristics, and descriptors are collectively referred to as *attributes*, and identified by UUIDs.
16 bits (e.g. "**180A**") or 128 bits (e.g. "**6BCF0ED3-68E3-4804-96D5-5AB8765FB9BC** ")

Application Profile Examples

- A2DP Advanced Audio Distribution Profile
- Global Navigation Satellite System Profile
- Hands-Free Profile
- Phone Book Access Profile
- SIM Access Profile
- Synchronization Profile
- Video Distribution Profile
- Blood Pressure Profile
- Cycling Power Profile
- Find Me Profile
- Heart Rate Profile
- Basic Printing Profile
- Dial-Up Networking Profile
- File Transfer Profile

Ref: Bluetooth SIG, "Adopted Bluetooth Profiles, Services, Protocols and Transports," https://www.bluetooth.org/en-us/specification/adopted-specifications

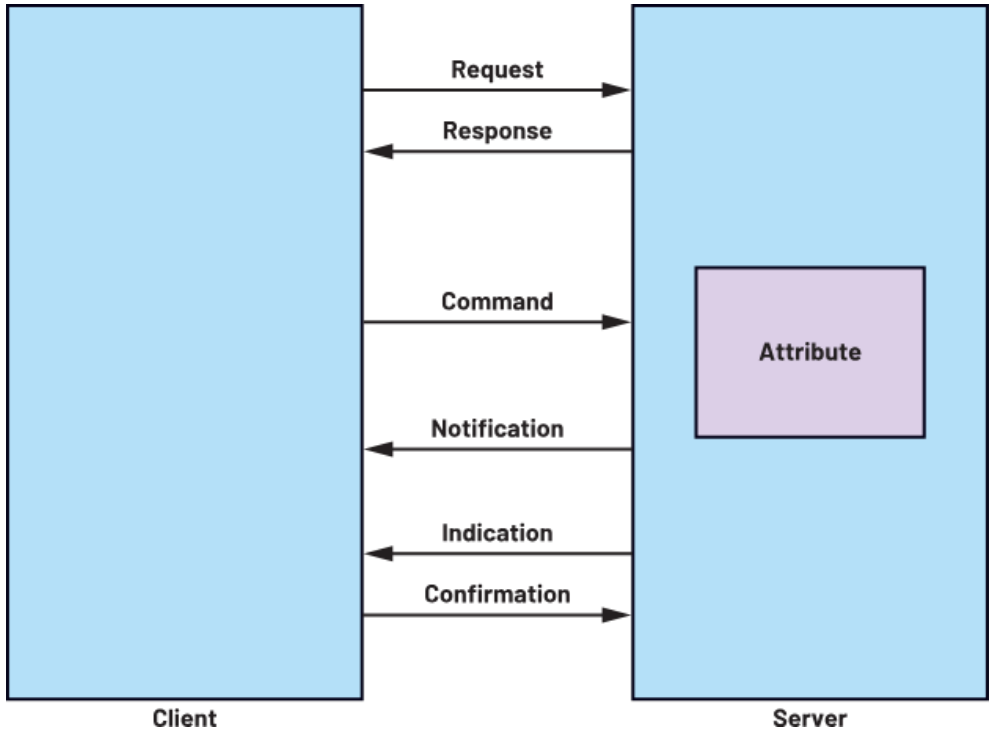
GATT Operations

There are 11 features defined in the GATT Profile

- Server Configuration
- Primary Service Discovery
- Relationship Discovery
- Characteristic Discovery
- Characteristic Descriptor Discovery
- Reading a Characteristic Value
- Writing a Characteristic Value
- Notification of a Characteristic Value
- Indication of a Characteristic Value
- Reading a Characteristic Descriptor
- Writing a Characteristic Descriptor

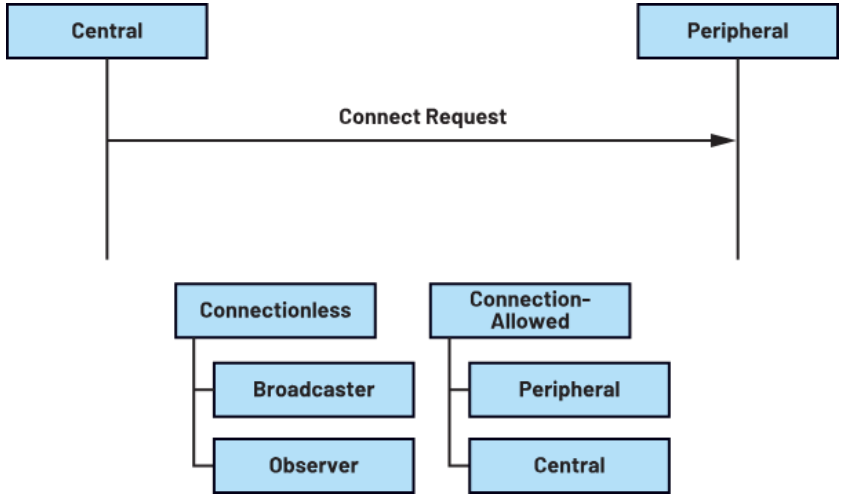
Ref: https://www.bluetooth.com/wp-content/uploads/Files/Specification/HTML/Core-54/out/en/host/generic-attribute-profile--gatt-.html (Section 4.2)

BLE Data Exchange (GATT)



A request PDU is sent by the client to the server, requesting a reply. A response PDU is the reply of the server to the client when a reply is requested.

The peripheral device, also broadcaster, advertises itself and awaits a connection request from a remote central device. The central device, acts as an observer, scanning for peripherals and initiating connection requests to the desired peripheral.



Ref: https://www.analog.com/en/resources/technical-articles/understanding-architecture-bluetooth-low-energy-stack.html

Bluetooth Beacons

- Advertising based on proximity
- Your phone broadcasts their presence if Bluetooth is turned on
- Primary aim of these broadcasts is to allow device discovery and indoor navigation
- Advertising frames consist of a header and max 27B of payload with multiple TLV-encoded data items
- May include signal strength ⇒ Distance
- Different Phones can send/receive Beacons
- Can be used for customized advertising, indoor location, geofencing

Bluetooth Smart Applications

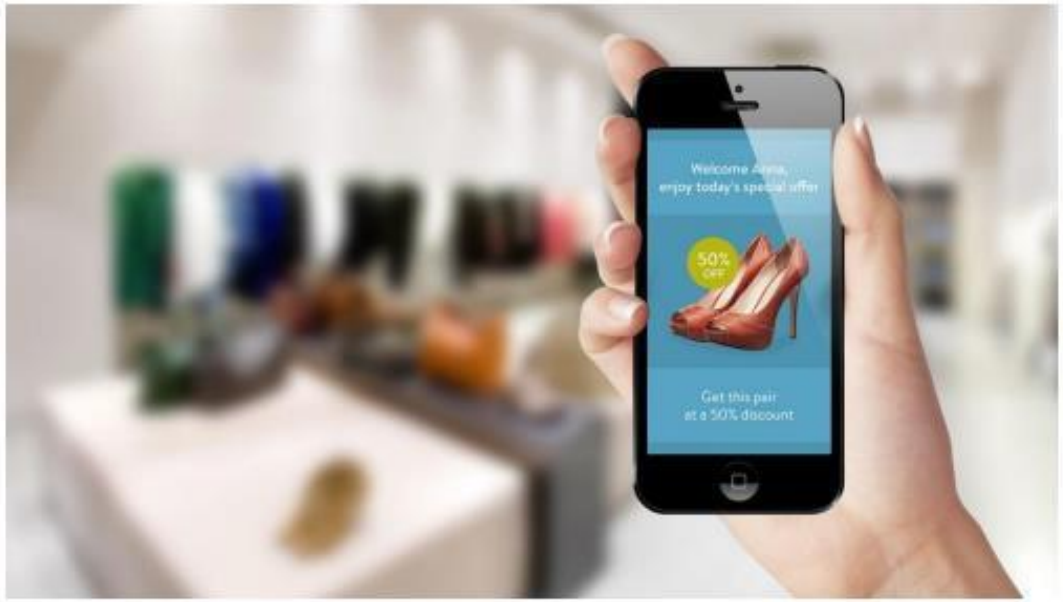
- Proximity: In car, In a Lectorial in SIT
- Locator: Keys, watches, Animals
- Health devices: Heart rate monitor, physical activities monitors, thermometer
- Sensors: Temperature, Battery Status, tire pressure
- Remote control: Open/close locks, turn on lights

Bluetooth Device Roles

Use Cases – Physical Security

Use Cases – Home Automation

Use Cases – Geo-fencing/ Positioning



Summary

1. Bluetooth basic rate uses frequency hoping over 79 1-MHz channels.
2. Has a fixed set of applications called "Profiles"
3. Bluetooth and Wi-Fi co-exist by time-sharing or adaptive frequency notching
4. Bluetooth Smart is designed for short broadcasts by sensors. 40 2-MHz channels with 3 channels reserved for advertising. One or two-message exchanges
5. Generic attribute profile allows new applications using UUID for data types

Do not edit
How to change the design

Audience Q&A

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